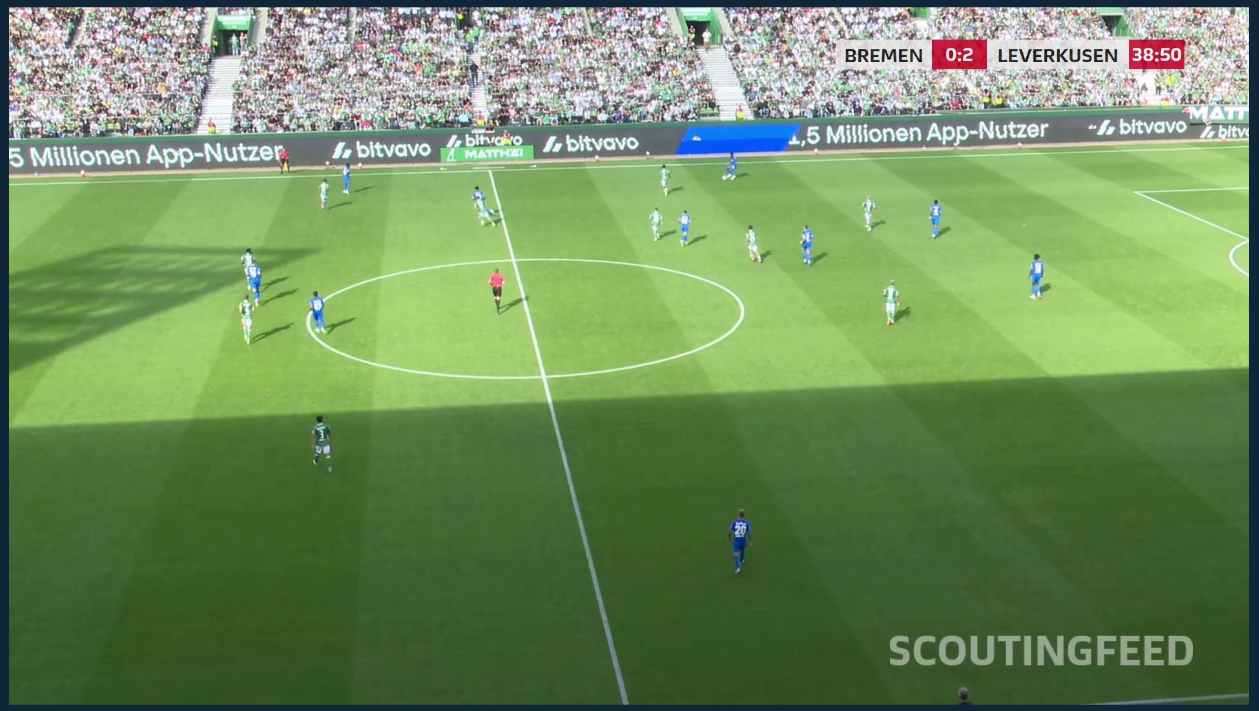
Visual preprocessing for robust football pose estimation

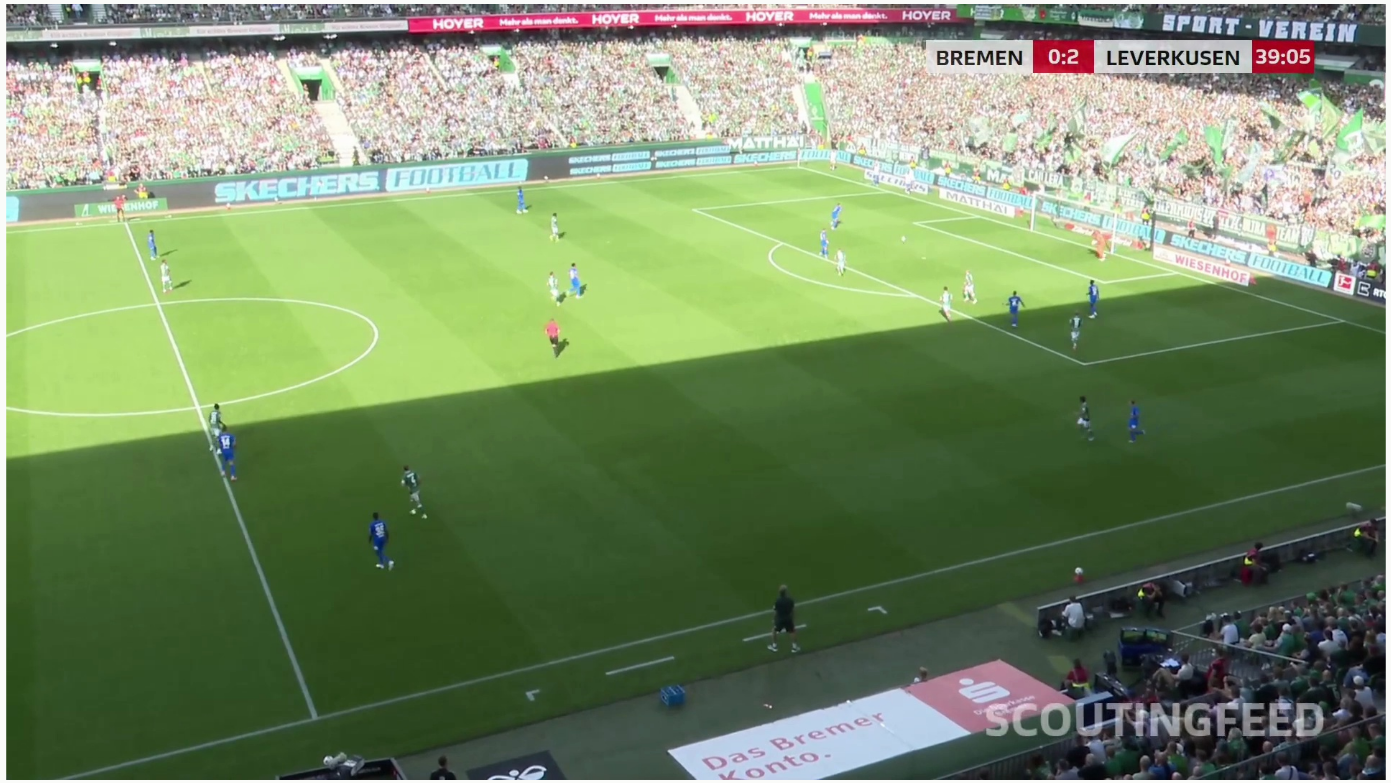
Single-camera scouting footage



Master’s thesis
Scope and current experiment status

Research question

Which preprocessing methods transfer to single-camera football pose estimation?



Independent variable

Processing method and its parameters

Measured outcomes

Detection coverage
Joint localization
Computational cost

Thesis scope

Single-camera inference, with independent reference labels for evaluation

Included

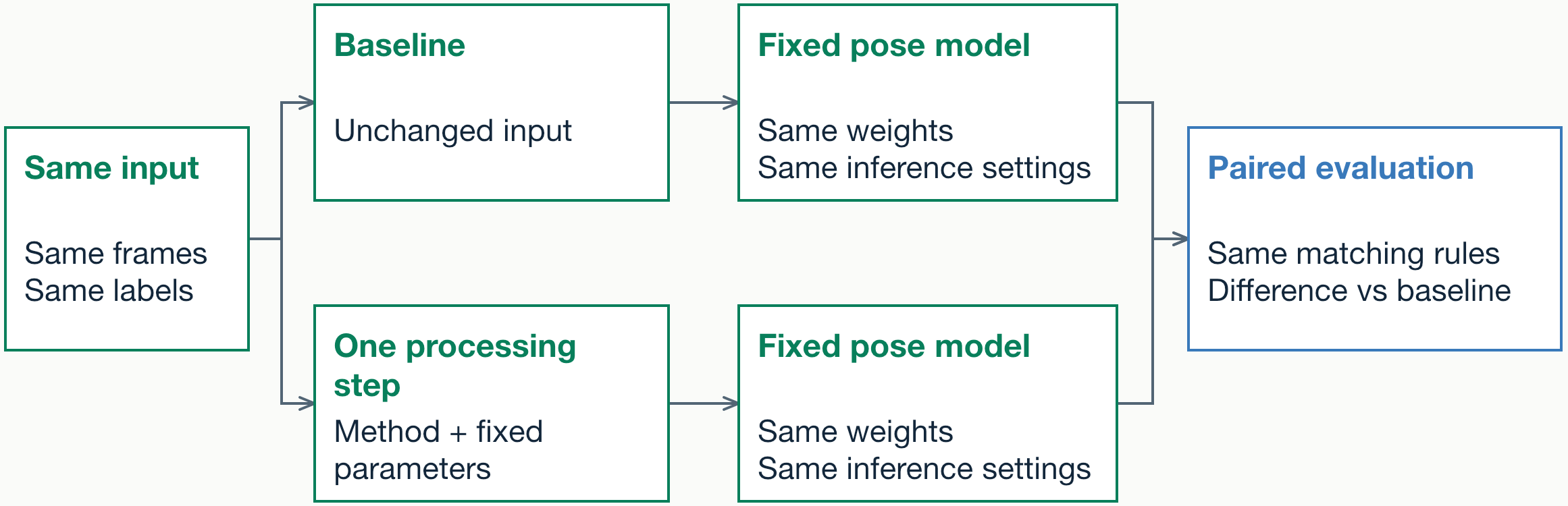
2D skeletal keypoints
Pre-recorded football video
Fixed pretrained pose models
Single steps, then combinations
Accuracy and computation cost

Outside the evaluation

New pose architectures or training
3D or multi-camera inference
Tracking, event or sensor fusion
Camera hardware experiments
Real-time deployment as a target

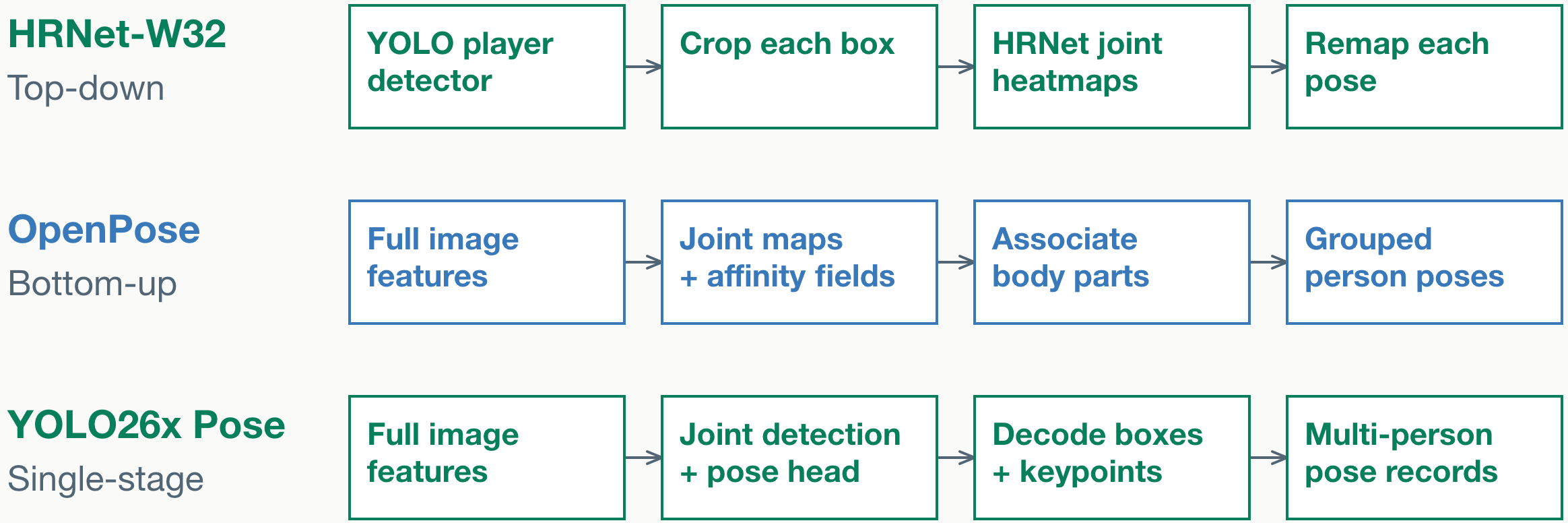
Controlled comparison

Fixed published pretrained checkpoints. No training or fine-tuning within the comparison.



Three routes to multi-person poses

“Multi-person” is an output capability shared by all three, not a separate architecture.



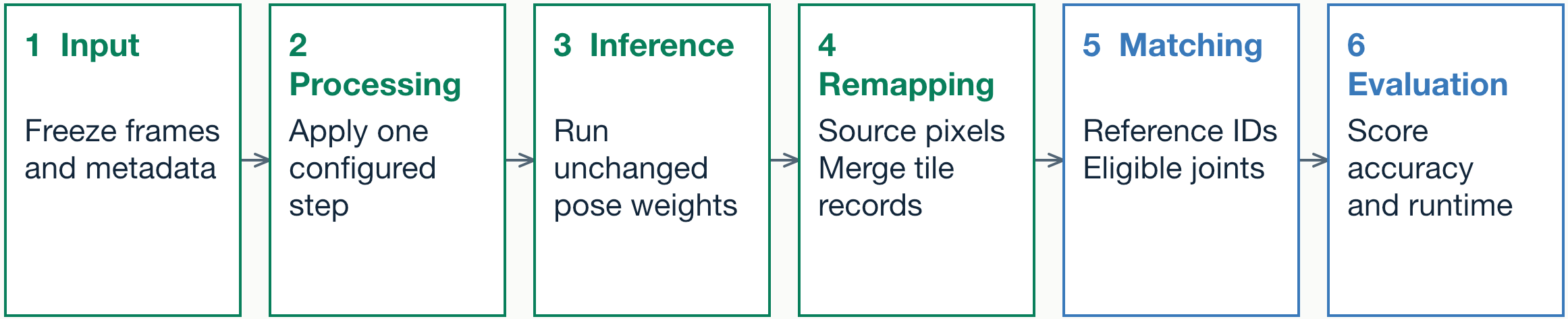
Evaluation data and ground truth

The exposé plans DFL/Werder joint labels for football and public datasets as references.

Role	Data and interpretation	Current position
Main input domain	Single wide-angle football camera Body joints in source-image pixels	Pilot: 782 frames One scouting clip
Football accuracy reference	DFL/Werder skeletal joint labels Align time, joint names and camera projection	Reference format and alignment pending
Public reference datasets	COCO Keypoints for model setup checks PoseTrack is an optional extension	COCO evaluator ready PoseTrack not integrated

Pipeline, step by step

Every prediction returns to the original frame before matching against ground truth.



Detection: precision and recall

First match predictions to eligible reference people, one-to-one, at a declared criterion.

Outcome	Meaning
True positive (TP)	A valid prediction matched to one reference person
False positive (FP)	An unmatched extra prediction, including a duplicate
False negative (FN)	A reference person with no valid matched prediction

Precision = TP / (TP + FP)

Illustration: TP = 8, FP = 2, FN = 4

8 / 10 = 80% correct predictions

Recall = TP / (TP + FN)

8 / 12 = 66.7% recovered targets

OKS: turn joint errors into a pose similarity

Object Keypoint Similarity: one score for a predicted pose and a reference person.

$OKS = (1 / N) \Sigma \exp[-d_i^2 / (2 A k_i^2)]$

Sum only over labelled reference joints ($v_i > 0$).

Symbol	How it enters the calculation
d_i	Euclidean distance from predicted joint i to its reference, in pixels
A	Ground-truth person area, in square pixels
$k_i = 2\sigma_i$	Fixed COCO joint-specific tolerance; not prediction confidence
N	Number of labelled reference joints contributing to this pose

Average Precision (AP) and Average Recall (AR)

COCO keypoints uses OKS, even where the evaluator prints the label “IoU”.



AP_t = mean interpolated precision at 101 recall levels

AP = mean AP_t for t = 0.50, 0.55, …, 0.95

AR = mean maximum recall over the same ten thresholds

AP50 / AP75 use one threshold; COCO retains up to 20 poses per image.

PCK: how many joints are close enough?

Percentage of Correct Keypoints; proposed scale: ground-truth person-box diagonal.

Normalized error: $e_i = \|p_i - g_i\|_2 / s$

PCK(α) = count(eligible joints with $e_i \le \alpha$) / N

N = all eligible GT joints; missing predictions count as incorrect.

Example: s = 200 px, α = 0.05 → tolerance = 10 px

Illustration: 8 eligible joints

6 within tolerance
PCK = 6 / 8 = 75%

7 confidently predicted
Coverage = 7 / 8 = 87.5%

Estimating the processing effect

Paired comparisons use the same reference population in every condition.

$$\Delta \text{ score} = \text{score(processed input)} - \text{score(baseline)}$$

1. **Fixed denominators**
   Count missed targets and unpredicted eligible reference joints
2. **Separate error sources**
   Report detection, pose localization and pitch-filter errors
3. **Independent confirmation**
   Choose settings on development clips, then test held-out sequences

Reproducibility safeguards

A result needs a recoverable input, configuration and execution environment.

Already recorded

Input and checkpoint SHA-256
Expanded YAML and processing order
Git revision and package versions
Hardware metadata and runner logs
Raw predictions and decision archives

Final-study verification

Pin actual container image digests
Freeze labels and evaluation settings
Repeat a reference run
Declare numerical tolerances
Measure cold and warm runtime separately

Common uses: contrast and denoising

Established applications motivate these transfer tests. Football benefits still need measurement.

Method	Common application	Question in football video
CLAHE	Medical / low-contrast imaging Local contrast enhancement	Does local contrast reveal player and joint detail?
Gamma darken / brighten	Photo / video exposure correction Nonlinear brightness adjustment	How sensitive are the models to brightness changes?
Non-local means	Computational photography Sensor-noise removal	Can noise reduction preserve the small body structures?
Bilateral filtering	Image enhancement and denoising Edge-preserving smoothing	Do cleaner regions and body edges help pose estimation?
Unsharp masking	Photo and print sharpening Local edge-contrast increase	Does sharpening help, or add halos and false structure?

Common uses: scale, regions and restoration

The current screen compares a baseline and eleven single processing conditions.

Method	Common application	Question in football video
Bicubic resize ×2	CV input sizing and resampling Non-learned enlargement	What does enlargement alone change in the pose pipeline?
FSRCNN ×2	Low-resolution photo enhancement Learned super-resolution	Does learned restoration add value beyond bicubic ×2?
Wiener deblur	Motion-blurred image restoration Assumed blur kernel	Can a fixed blur model help with camera or player motion?
Tiling	Aerial / satellite detection Small objects in large images	Do larger effective player scales outweigh boundary errors?
Player crops	Top-down pose estimation Region-of-interest analysis	Does person-focused input help despite detector dependence?

Current experiment status

Exploratory detection yield on the football clip. Accuracy evaluation remains pending.

Model	Baseline on pitch	Example condition	On pitch
YOLO26	357	Bilateral denoising	2,115
OpenPose	11,647	CLAHE	15,338
HRNet	3,602	Bicubic ×2	15,554

Completed: 35 of 36 model jobs

Open cases: HRNet tiling, YOLO cropping and the failed OpenPose crop job

Progress against the exposé

The processing system and pilot exist. The central football accuracy study remains ahead.

Work package	Status	Evidence or remaining requirement
Modular processing pipeline	Implemented	Canonical predictions, provenance Automated reports and rendered videos
Single-step football pilot	35 / 36 jobs complete	12 conditions × 3 models on one clip Crop and tiling anomalies remain
Public model checks	COCO evaluator integrated	Historical YOLOv8x / OpenPose results exist YOLO26 confirmation remains pending
Football mAP / PCK scoring	Blocked on reference data	DFL joint format, camera projection and timestamp synchronization
Combinations and guidelines	Planned	Score against independent labels Confirm on held-out video sequences

Remaining thesis work

1. **Football reference-data contract**

   Confirm DFL joints, timestamps, camera projection and access

2. **Accuracy evaluation and runner checks**

   Freeze matching rules and resolve crop / tiling anomalies

3. **Controlled single steps and combinations**

   Use held-out clips and compare accuracy with computational cost

4. **Thesis conclusions and practical guidelines**

   Explain which methods transfer, for which model and under which conditions

Football reference-data protocol

Proposed workflow before the final accuracy experiment

1. **Define the evaluation sample**

   Include varied scale, lighting, occlusion and camera views

2. **Align the reference to the video**

   Match timestamps, player identities and joint definitions

3. **Review labels independently**

   Record visibility, uncertainty and annotation disagreements

4. **Freeze development and test sets**

   Separate video sequences and version the approved labels

Experimental matrix

One unchanged baseline and eleven individual processing conditions

Processing family	Conditions
Scale and regions	Bicubic ×2, FSRCNN ×2, 2 × 2 tiling, detector-guided crops
Contrast and brightness	CLAHE, gamma darken, gamma brighten
Noise and restoration	Non-local means, bilateral denoising, unsharp, Wiener deblur

YOLO26x Pose
OpenPose BODY_25
HRNet-W32

Current football pilot
12 conditions × 3 models
782 identical source frames

Backup: on-pitch screening overview 1/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Baseline	357 (reference)	11,647 (reference)	3,602 (reference)
Resize ×2	283 (-20.7%)	10,934 (-6.1%)	15,554 (+331.8%)
CLAHE	1,156 (+223.8%)	15,338 (+31.7%)	3,995 (+10.9%)
Gamma darken	315 (-11.8%)	12,193 (+4.7%)	3,634 (+0.9%)
Gamma brighten	440 (+23.2%)	11,029 (-5.3%)	3,568 (-0.9%)
NLM denoising	615 (+72.3%)	8,054 (-30.8%)	3,037 (-15.7%)

Backup: on-pitch screening overview 2/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Bilateral denoising	2,115 (+492.4%)	12,581 (+8.0%)	3,216 (-10.7%)
Unsharp	491 (+37.5%)	11,860 (+1.8%)	2,675 (-25.7%)
Motion deblur	1,627 (+355.7%)	7,362 (-36.8%)	214 (-94.1%)
Super-resolution ×2	312 (-12.6%)	11,583 (-0.5%)	15,565 (+332.1%)
Tiling	3,402 (+852.9%)	16,072 (+38.0%)	1 (-100.0%)
Detector crop	0 (-100.0%)	FAILED	3,467 (-3.7%)

Backup: PCP — Percentage of Correct Parts

Proposed strict-endpoint convention; other PCP variants must not be mixed with it.

Reference limb length: $L = \|g(a) - g(b)\|_2$

Correct limb if BOTH endpoint errors $\leq \alpha L$

PCP = correct eligible limbs / all eligible reference limbs

Illustration: $L = 40$ px and $\alpha = 0.5 \rightarrow 20$ px tolerance

Endpoint errors 12 px and 18 px: pass
Endpoint errors 12 px and 22 px: fail

Backup: PCK-AUC summarizes tolerances

AUC here is the area under PCK versus distance threshold, not the precision-recall curve.

$$\text{Normalized AUC} = (1 / \alpha_{max}) \times \int_{0}^{\alpha_{max}} \text{PCK}(\alpha)\, d\alpha$$

Threshold α	0.00	0.05	0.10
Illustrative PCK	0.00	0.50	1.00

Trapezoid estimate over [0, 0.10]

$$\text{AUC} \approx [0.05 \times (0 + 0.50)/2 + 0.05 \times (0.50 + 1)/2] / 0.10$$

$$= 0.50 \quad (50\% \text{ of the maximum possible area})$$

Backup: proxy metrics and runtime

The exposé proposes complementary diagnostics alongside ground-truth pose accuracy.

Mean confidence = $\Sigma\ c_i$ / K valid predicted joints

Joint coverage(τ) = count(matched eligible joints with $c_i \geq \tau$) / N

Diagnostic	Calculation or interpretation
Yield	Person-pose records / source frames is already reported It differs from keypoint count and detection accuracy
Temporal stability	Planned motion-aware joint-jitter diagnostic Needs consistent short-term associations and camera-motion handling
Runtime	Preprocessing, inference and total wall time Report cold materialization and warm-cache timings separately

Backup: AP from a confidence-ranked list

Illustration at one OKS threshold: 3 reference people, 5 predictions, no ignored cases.

Rank / score	Outcome	Cum. TP / FP	Precision	Recall
1 / 0.95	TP	1 / 0	1.000	0.333
2 / 0.90	FP	1 / 1	0.500	0.333
3 / 0.80	TP	2 / 1	0.667	0.667
4 / 0.70	TP	3 / 1	0.750	1.000
5 / 0.60	FP	3 / 2	0.600	1.000

Precision envelope: 1.00 up to recall 1/3; then 0.75

$AP_t = (34 \times 1.00 + 67 \times 0.75) / 101 = 0.834$; $R_{max,t} = 1.00$

Backup: OKS scale and visibility example

Illustrative tolerances below are not the COCO sigma values for a particular named joint.

Joint error d	GT area A	Tolerance k	Contribution $\exp[-d^2/(2Ak^2)]$
0 px	10,000 px^2	0.10	1.000
10 px	10,000 px^2	0.10	0.607
20 px	10,000 px^2	0.10	0.135

For these 3 labelled joints: OKS = (1 + 0.607 + 0.135) / 3 = 0.581

GT visibility: 0 = unlabelled; 1 = labelled, occluded; 2 = labelled, visible.
Both 1 and 2 contribute. Predicted confidence is not a visibility label.

Backup: architecture-specific bottlenecks

All methods share an output task; their internal representations and failure modes differ.

Pipeline	Internal mechanism	Relevant limitation
HRNet top-down	Parallel high/low-resolution branches Repeated fusion → joint heatmaps	Missed detector boxes give no pose Pose work scales with crop count
OpenPose bottom-up	Joint maps + part-affinity vectors Group connected limbs into people	Tiny or occluded joint evidence Ambiguous grouping in crowds
YOLO26 single-stage	Shared image features Person boxes + keypoint predictions	Missed instances and localization Depend on scale and image features

Our comparison maps BODY_25 to the shared COCO17 joint schema.
A fixed input-size setting does not mean the three networks see identical representations.

Backup: evaluation protocol decisions

Official COCO scoring exists. Football-specific accuracy conventions still need agreement.

Decision	Rule to document
Reference data	DFL body-joint schema, source-image projection, timestamps and visibility
Matching and targets	Same reference people, one-to-one matching and duplicate handling
PCK / PCP	Scale, threshold, limb list and eligible-joint aggregation
AP / AR	OKS conventions, person-score ranking and retained-prediction limit
Supplementary diagnostics	Confidence calibration, coverage and motion-aware temporal stability
Statistical comparison	Paired sequence-level effects, held-out clips and uncertainty intervals